

A Weight-5 Apéry-like Sequence $S_{20}(n)$: Formally Verified Supercongruences and a Conjectural Calabi–Yau Period (preprint, v6 — under community review)

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Abstract

We study the integer sequence

$$S_{20}(n) = \sum_{k=0}^n \binom{n}{k}^4 \binom{n+k}{k} = 1, 3, 55, 1155, 29751, 852753, \dots,$$

a weight-5 (more precisely weight-(4, 1)) Apéry-like binomial sum that we have been unable to locate in the literature or in the OEIS. A draft OEIS submission (proposed identifier [A397213](#)) is *pending editorial review and has not been approved*; we therefore make no novelty claim beyond “we did not find it.” The purpose of this note is narrow and, we hope, honest: we give complete, human-readable proofs of three arithmetic congruences satisfied by S_{20} , and we report that *each of these three results has also been verified in the Lean 4 proof assistant with no `sorry`, no `admit`, and no axioms beyond Lean’s three standard foundations (propext, Classical.choice, Quot.sound)*. The three results are:

- (I) the *cubic supercongruence* $S_{20}(p) \equiv 3 \pmod{p^3}$ for every prime $p \geq 5$;
- (II) *Wolstenholme’s theorem* $\binom{2p}{p} \equiv 2 \pmod{p^3}$ for $p \geq 5$, reproved from first principles (it is the one external input of (I));
- (III) the *Apéry-style boundary congruence* $S_{20}(p-1) \equiv 1 \pmod{p}$ for every prime p .

We are explicit about what is *not* settled: the order-4 minimal linear recurrence is computed but not formally certified for general n ; the mod- p^3 refinement of (III) is an open conjecture; and the interpretation of S_{20} as a Calabi–Yau period is a structural analogy, not a theorem. We invite correction, refutation, and independent verification.

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1 Introduction and honest scope

Apéry's irrationality proofs for $\zeta(2)$ and $\zeta(3)$ rest on integer sequences satisfying three-term recurrences with remarkable arithmetic properties. A now-classical theme — developed by Beukers, Zagier, Almkvist–Zudilin, and others — is that such sequences often coincide with periods of one-parameter families of Calabi–Yau manifolds, and that their p -adic behaviour (supercongruences) reflects this geometry.

This note concerns the specific binomial sum

$$S_{20}(n) = \sum_{k=0}^n \binom{n}{k}^4 \binom{n+k}{k}. \quad (1)$$

It is the $A = 4$, $B = 1$ member of the family $\sum_k \binom{n}{k}^A \binom{n+k}{k}^B$; the Apéry numbers for $\zeta(3)$ are $A = B = 2$.

What this paper claims, and what it does not. We have tried to separate rigour from speculation cleanly. The reader should read the following table as the precise scope of the work.

Statement	Status	Where
First terms / closed form (1)	verified (exact arithmetic)	§2
$S_{20}(p) \equiv 3 \pmod{p^3}$, $p \geq 5$	proved (+ Lean)	§3
$\binom{2p}{p} \equiv 2 \pmod{p^3}$, $p \geq 5$	proved (+ Lean)	§4
$S_{20}(p-1) \equiv 1 \pmod{p}$, all p	proved (+ Lean)	§5
$S_{20}(p-1) \equiv 1 \pmod{p^3}$	open conjecture (numeric only)	§5
order-4 recurrence (general n)	computed, not formally certified	§7
Calabi–Yau period interpretation	conjectural analogy	§7

On the Lean 4 formalization. All three congruence theorems below were transcribed into Lean 4 (Mathlib, toolchain v4.31.0). Building the development with `lake build` succeeds with no errors; the command `#print axioms` applied to each final theorem returns only `[proptext, Classical.choice, Quot.sound]`. We stress what this does and does not mean: it means the *stated* theorems are correct relative to Lean's kernel and Mathlib; it does *not* certify the recurrence for general n , nor any of the conjectural material. We give the human proofs in full here precisely so that they can be checked by a mathematician without trusting the machine. We would be grateful for scrutiny of both.

2 Preliminaries

Throughout, p denotes a prime and $\binom{n}{k}$ the usual binomial coefficient, with $\binom{n}{k} = 0$ for $k > n$. We write the summand of (1) as

$$T(n, k) = \binom{n}{k}^4 \binom{n+k}{k}, \quad S_{20}(n) = \sum_{k=0}^n T(n, k).$$

We use two elementary facts repeatedly.

Lemma 2.1 (Prime divides interior binomials). *If p is prime and $1 \leq k \leq p-1$, then $p \mid \binom{p}{k}$.*

Proof. From $k \binom{p}{k} = p \binom{p-1}{k-1}$ we get $p \mid k \binom{p}{k}$. Since $1 \leq k \leq p-1$, the prime p does not divide k , hence $p \mid \binom{p}{k}$. (This is `Nat.Prime.dvd_choose_self` in Mathlib.) $\square$

Lemma 2.2 (Factorial p -divisibility). *For a prime p and $m \in \mathbb{N}$, one has $p \mid m!$ if and only if $p \leq m$.*

Proof. If $p \leq m$ then p is one of the factors of $m!$. Conversely, if $p \mid m!$ then, as p is prime, p divides one of $1, 2, \dots, m$, forcing $p \leq m$. (This is `Nat.Prime.dvd_factorial`.) $\square$

3 The cubic supercongruence $S_{20}(p) \equiv 3 \pmod{p^3}$

The main arithmetic result of this note is the following.

Theorem 3.1 (Cubic supercongruence). *For every prime $p \geq 5$,*

$$S_{20}(p) \equiv 3 \pmod{p^3}.$$

The proof has two ingredients: an elementary *collapse* of the sum to its two boundary terms modulo p^3 (Proposition 3.2), and the classical Wolstenholme congruence for $\binom{2p}{p}$ (Theorem 4.5, proved independently in §4). The collapse is the part specific to S_{20} , and it is genuinely simple; the arithmetic depth is entirely in Wolstenholme.

3.1 The collapse to boundary terms

Proposition 3.2 (Collapse). *For every prime p ,*

$$S_{20}(p) \equiv 1 + \binom{2p}{p} \pmod{p^3}.$$

Proof. Split the sum (1) at its endpoints:

$$S_{20}(p) = T(p, 0) + \sum_{k=1}^{p-1} T(p, k) + T(p, p).$$

The boundary terms are explicit:

$$T(p, 0) = \binom{p}{0}^4 \binom{p}{0} = 1, \quad T(p, p) = \binom{p}{p}^4 \binom{2p}{p} = \binom{2p}{p}.$$

It remains to show that each interior term vanishes modulo p^3 . Fix $1 \leq k \leq p-1$. By Lemma 2.1, $p \mid \binom{p}{k}$, hence $p^4 \mid \binom{p}{k}^4$. The companion factor $\binom{p+k}{k}$ is an integer, so

$$p^4 \mid \binom{p}{k}^4 \binom{p+k}{k} = T(p, k),$$

and a fortiori $p^3 \mid T(p, k)$. Summing, $p^3 \mid \sum_{k=1}^{p-1} T(p, k)$, which gives the claim. $\square$

Remark 3.3. The factor of $\binom{p}{k}^4$ is crucial: a single power would give only $p^1 \mid \binom{p}{k}$, enough for a mod- p statement but not mod- p^3 . The fourth power provides p^4 “for free,” so the companion binomial $\binom{p+k}{k}$ — whatever its p -adic valuation — cannot spoil divisibility by p^3 . This is the only place the exponent 4 in (1) is used in this section.

3.2 Proof of Theorem 3.1

Proof. By Proposition 3.2, $S_{20}(p) \equiv 1 + \binom{2p}{p} \pmod{p^3}$. By Wolstenholme's theorem (Theorem 4.5), valid for $p \geq 5$, we have $\binom{2p}{p} \equiv 2 \pmod{p^3}$. Therefore

$$S_{20}(p) \equiv 1 + 2 = 3 \pmod{p^3}. \quad \square$$

Corollary 3.4 (Unconditional mod p^2). *For every prime p , $S_{20}(p) \equiv 3 \pmod{p^2}$.*

Proof. The collapse (Proposition 3.2) holds mod p^3 and hence mod p^2 . Babbage's congruence $\binom{2p}{p} \equiv 2 \pmod{p^2}$ holds for all primes (Theorem 4.2), so $S_{20}(p) \equiv 1 + 2 = 3 \pmod{p^2}$. No lower bound on p is needed here. $\square$

4 Wolstenholme's theorem from first principles

For completeness and because our Lean development does not assume it, we prove the two congruences for the central binomial coefficient used above. Nothing here is new — these are classical results of Babbage (1819) and Wolstenholme (1862) — but we give self-contained arguments matching the formalization.

The starting point is a Vandermonde identity.

Lemma 4.1 (Central binomial as a sum of squares). *For all $p \geq 0$, $\binom{2p}{p} = \sum_{k=0}^p \binom{p}{k}^2$.*

Proof. Chu–Vandermonde gives $\binom{2p}{p} = \sum_{k=0}^p \binom{p}{k} \binom{p}{p-k}$, and $\binom{p}{p-k} = \binom{p}{k}$. $\square$

Theorem 4.2 (Babbage, mod p^2). *For every prime p , $\binom{2p}{p} \equiv 2 \pmod{p^2}$.*

Proof. By Lemma 4.1, $\binom{2p}{p} = \binom{p}{0}^2 + \sum_{k=1}^{p-1} \binom{p}{k}^2 + \binom{p}{p}^2 = 2 + \sum_{k=1}^{p-1} \binom{p}{k}^2$. For $1 \leq k \leq p-1$, Lemma 2.1 gives $p \mid \binom{p}{k}$, hence $p^2 \mid \binom{p}{k}^2$. Thus the interior sum is divisible by p^2 and $\binom{2p}{p} \equiv 2 \pmod{p^2}$. $\square$

To upgrade to mod p^3 we need the harmonic input. Work in the field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$.

Lemma 4.3 (Pascal residues). *For a prime p and $0 \leq j \leq p-1$, $\binom{p-1}{j} \equiv (-1)^j \pmod{p}$.*

Proof. Induction on j . For $j = 0$, $\binom{p-1}{0} = 1$. For the step, Pascal's rule gives $\binom{p}{j+1} = \binom{p-1}{j+1} + \binom{p-1}{j}$. By Lemma 2.1, $p \mid \binom{p}{j+1}$ when $1 \leq j+1 \leq p-1$, so in $\mathbb{F}_p$ we have $\binom{p-1}{j+1} \equiv -\binom{p-1}{j} \equiv -(-1)^j = (-1)^{j+1}$. $\square$

Lemma 4.4 (Harmonic core). *For a prime $p \geq 5$, $\sum_{k=1}^{p-1} \frac{1}{k^2} \equiv 0 \pmod{p}$, where $1/k$ denotes the inverse of k in $\mathbb{F}_p$.*

Proof. As k ranges over $1, \dots, p-1$, the inverses k^{-1} range over the same set $\mathbb{F}_p^\times$, so $\sum_{k=1}^{p-1} k^{-2} = \sum_{x \in \mathbb{F}_p^\times} x^2 = \sum_{x \in \mathbb{F}_p} x^2$ (the $x = 0$ term is 0). For any finite field $\mathbb{F}_q$ and exponent i with $0 < i < q-1$, one has $\sum_{x \in \mathbb{F}_q} x^i = 0$ (a standard fact: the sum of all i -th powers over a finite field vanishes unless $(q-1) \mid i$). Here $q = p$, $i = 2$, and $2 < p-1$ exactly when $p \geq 5$, so the sum is 0. $\square$

Theorem 4.5 (Wolstenholme, mod p^3). *For every prime $p \geq 5$, $\binom{2p}{p} \equiv 2 \pmod{p^3}$.*

Proof. By Lemma 4.1 and the boundary evaluation as in Theorem 4.2,

$$\binom{2p}{p} - 2 = \sum_{k=1}^{p-1} \binom{p}{k}^2.$$

For $1 \leq k \leq p-1$ write $\binom{p}{k} = \frac{p}{k} \binom{p-1}{k-1}$, which is the integer identity $k \binom{p}{k} = p \binom{p-1}{k-1}$ solved for $\binom{p}{k}$; set $d_k := \binom{p}{k}/p = \frac{1}{k} \binom{p-1}{k-1} \in \mathbb{Z}$ (an integer by Lemma 2.1). Then $\binom{p}{k}^2 = p^2 d_k^2$ and

$$\binom{2p}{p} - 2 = p^2 \sum_{k=1}^{p-1} d_k^2.$$

Thus $\binom{2p}{p} \equiv 2 \pmod{p^3}$ if and only if $p \mid \sum_{k=1}^{p-1} d_k^2$. Reduce d_k modulo p . From $k d_k = \binom{p-1}{k-1}$ and Lemma 4.3, $k d_k \equiv (-1)^{k-1} \pmod{p}$, so in $\mathbb{F}_p$

$$d_k \equiv (-1)^{k-1} k^{-1}, \quad d_k^2 \equiv k^{-2}.$$

Hence $\sum_{k=1}^{p-1} d_k^2 \equiv \sum_{k=1}^{p-1} k^{-2} \equiv 0 \pmod{p}$ by Lemma 4.4 (using $p \geq 5$). Therefore $p \mid \sum_{k=1}^{p-1} d_k^2$ and $\binom{2p}{p} \equiv 2 \pmod{p^3}$. $\square$

Remark 4.6. The hypothesis $p \geq 5$ enters in exactly one place: Lemma 4.4 needs $2 < p-1$. For $p=3$ one checks directly $\binom{6}{3} = 20 \equiv 2 \pmod{9}$ but $20 \not\equiv 2 \pmod{27}$, so the mod- p^3 statement genuinely fails at $p=3$; the mod- p^2 statement (Theorem 4.2) survives.

5 The Apéry-style boundary congruence

The second sequence-specific result concerns the value at $p-1$.

Theorem 5.1 (Apéry-style congruence mod p). *For every prime p , $S_{20}(p-1) \equiv 1 \pmod{p}$.*

The mechanism mirrors the collapse of §3, but now the divisibility comes from the *companion* binomial rather than from $\binom{p-1}{k}$.

Lemma 5.2 (Companion divisibility). *For a prime p and $1 \leq k \leq p-1$, $p \mid \binom{p-1+k}{k}$.*

Proof. The factorial identity for binomial coefficients gives

$$\binom{p-1+k}{k} \cdot k! \cdot (p-1)! = (p-1+k)!,$$

since $(p-1+k)-k = p-1$. As $1 \leq k \leq p-1$ we have $p \leq p-1+k$, so by Lemma 2.2 the prime p divides the right-hand side $(p-1+k)!$. On the other hand $k < p$ and $p-1 < p$, so by Lemma 2.2 again $p \nmid k!$ and $p \nmid (p-1)!$. Since p is prime and divides the product $\binom{p-1+k}{k} k! (p-1)!$ but neither $k!$ nor $(p-1)!$, it must divide the remaining factor $\binom{p-1+k}{k}$. $\square$

Proof of Theorem 5.1. Split the sum at $k=0$:

$$S_{20}(p-1) = T(p-1, 0) + \sum_{k=1}^{p-1} T(p-1, k), \quad T(p-1, 0) = \binom{p-1}{0}^4 \binom{p-1}{0} = 1.$$

For each interior index $1 \leq k \leq p-1$, $T(p-1, k) = \binom{p-1}{k}^4 \binom{p-1+k}{k}$, and Lemma 5.2 gives $p \mid \binom{p-1+k}{k}$, hence $p \mid T(p-1, k)$. Therefore $\sum_{k=1}^{p-1} T(p-1, k) \equiv 0 \pmod{p}$ and $S_{20}(p-1) \equiv 1 \pmod{p}$. $\square$

Conjecture 5.3 (Strong Apéry-style congruence). For every prime $p \geq 5$, $S_{20}(p-1) \equiv 1 \pmod{p^3}$.

We have verified Conjecture 5.3 for all primes $5 \leq p \leq 200$ (see the reproducibility report). We do *not* have a proof; the mod- p argument above does not extend naively, because $\binom{p-1}{k}^4 \equiv 1 \pmod{p}$ controls only the leading p -adic order. A proof would presumably require a harmonic-sum analysis in the spirit of §4 (likely involving $\sum 1/k$ and $\sum 1/k^2$ to higher order), and we would welcome one.

6 Numerical and formal verification

All numerical claims are reproduced by a single self-contained script, `python/verify_all.py` (standard library only), whose checks include: the first 18 terms against an independent transcription; the cubic supercongruence and Wolstenholme/Babbage congruences for all primes up to 200; the Apéry-style congruence mod p (and the conjectural mod p^3) up to 200; the collapse identity of Proposition 3.2; and the diagonal evaluation discussed in §7. The script terminates with exit code 0 and the message “*ALL NUMERICAL CHECKS PASSED.*”

Theorems 3.1, 4.5 (with Theorem 4.2), and 5.1 have been formalized in Lean 4 (Mathlib, toolchain v4.31.0) as, respectively, `Supercongruence.supercongruence_unconditional`, `Wolstenholme.wolstenholme / Wolstenholme.babbage`, and `ApéryCongruence.apéry_congruence`. Each compiles under `lake build` and, under `#print axioms`, depends only on `propxt`, `Classical.choice`, and `Quot.sound`. The full audit is recorded in `VERIFICATION_REPORT.md`.

7 What remains open

We close by stating clearly the parts of the broader picture that are *not* theorems in this paper, in the hope that experts will engage with them.

Linear recurrence. Numerical linear algebra (a $\mathbb{Q}$ -nullspace computation) indicates that $S_{20}(n)$ satisfies a minimal linear recurrence of order 4 and polynomial degree 13, with an order-5, degree-9 left-multiple. We have verified small base cases in Lean, but the general- n statement currently rests on a creative-telescoping (Zeilberger) certificate that we have *not* re-derived inside a formally verified setting. A verified certificate would be a valuable contribution.

Diagonal representation. We observe the identity, verified for $0 \leq n \leq 6$ and provable by elementary power-series extraction,

$$S_{20}(n) = \text{Diag} \left[\frac{1}{(1-x_1)(1-x_2)(1-x_3)(1-x_4)(1-x_5) - x_1x_2x_3x_4} \right],$$

i.e. $S_{20}(n)$ is the coefficient of $(x_1 \cdots x_5)^n$. Indeed expanding $1/(D-M) = \sum_{r \geq 0} M^r / D^{r+1}$ with $D = \prod_i (1-x_i)$, $M = x_1x_2x_3x_4$ and extracting the diagonal yields exactly $\sum_r \binom{n}{r}^4 \binom{n+r}{r}$. The *symmetric* five-variable candidate does not represent S_{20} ; its diagonal is 2^n .

Calabi–Yau period. The order of the (conjectural) minimal Picard–Fuchs operator is 4, which would be consistent with a one-parameter family whose period geometry is that of a Calabi–Yau 3-fold. We computed mirror-map coefficients q_d via exact rational arithmetic and found them to be integers for $1 \leq d \leq 16$, consistent with Lian–Yau integrality. We emphasize that this is *evidence and analogy*: we do not prove that S_{20} is the period of a specific Calabi–Yau manifold, and the precise geometry (including its dimension) deserves expert scrutiny.

Invitation. This is a preprint released specifically to invite review. Corrections, counterexamples, a proof of Conjecture 5.3, a verified recurrence certificate, an identification (or refutation) of the Calabi–Yau interpretation, or a prior appearance of S_{20} in the literature would all be very welcome. Please open an issue or pull request at the repository.

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References

- [1] R. Apéry, *Irrationalité de $\zeta(2)$ et $\zeta(3)$* , Astérisque **61** (1979), 11–13.
- [2] C. Babbage, *Demonstration of a theorem relating to prime numbers*, Edinburgh Phil. J. **1** (1819), 46–49.
- [3] J. Wolstenholme, *On certain properties of prime numbers*, Quart. J. Pure Appl. Math. **5** (1862), 35–39.
- [4] B. H. Lian and S.-T. Yau, *Arithmetic properties of mirror map and quantum coupling*, Comm. Math. Phys. **176** (1996), 163–191.
- [5] G. Almkvist, C. van Enckevort, D. van Straten, W. Zudilin, *Tables of Calabi–Yau equations*, arXiv:math/0507430.
- [6] The mathlib Community, *The Lean mathematical library*, CPP 2020.
- [7] X. Callens, *Mirror Map Sieve* (software and Lean formalization), <https://github.com/xaviercallens/Mirror-Map-Sieve>, DOI: 10.5281/zenodo.20747943, 2026.